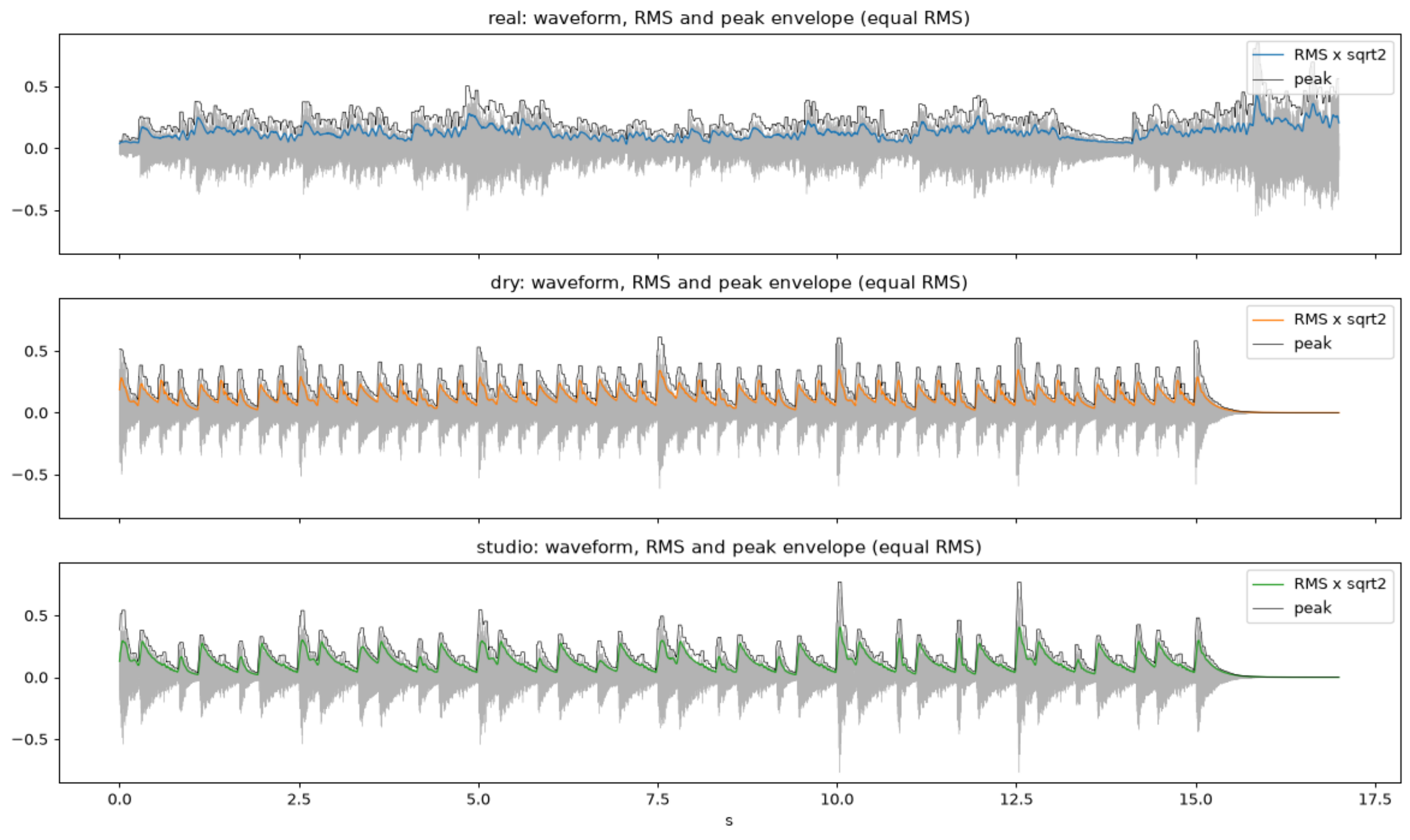
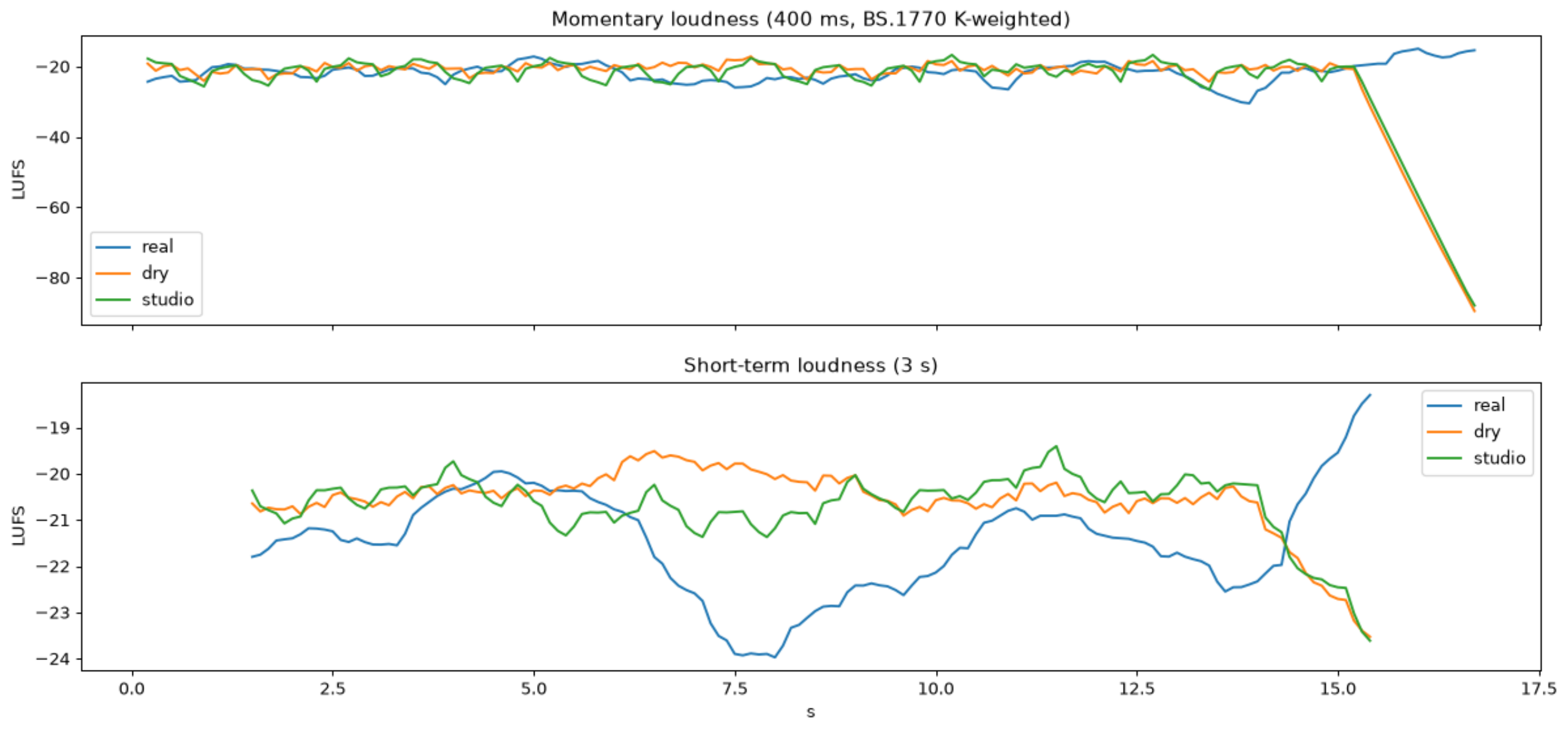
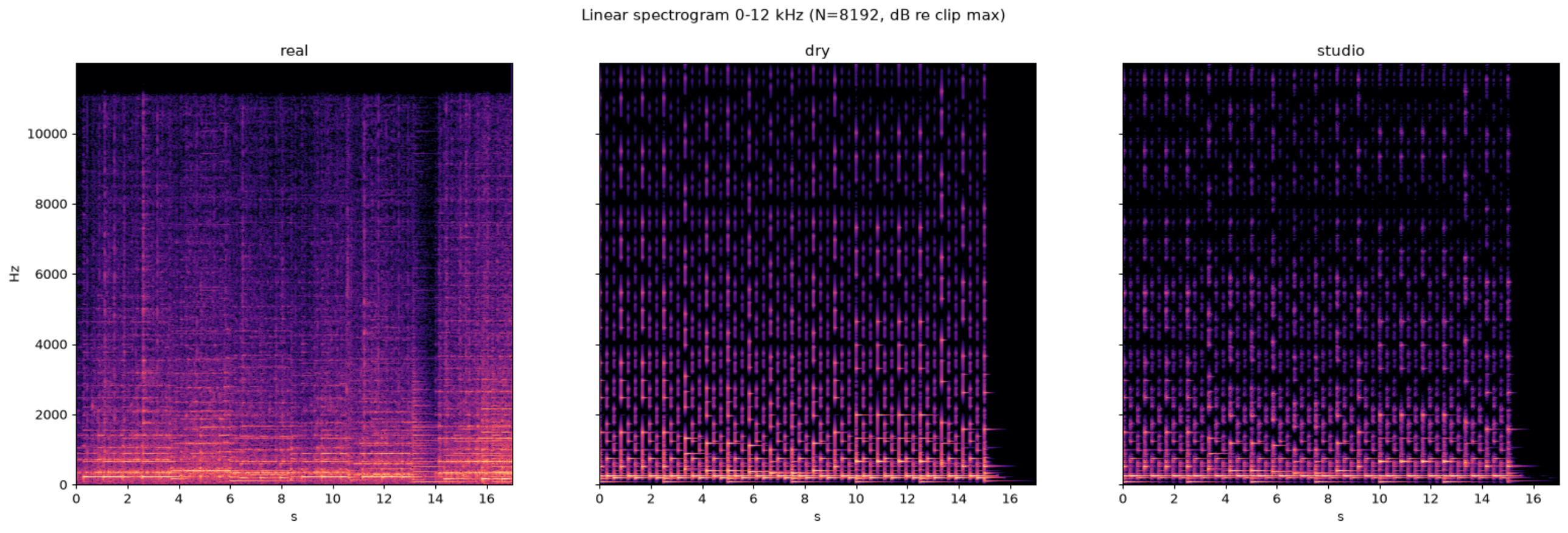
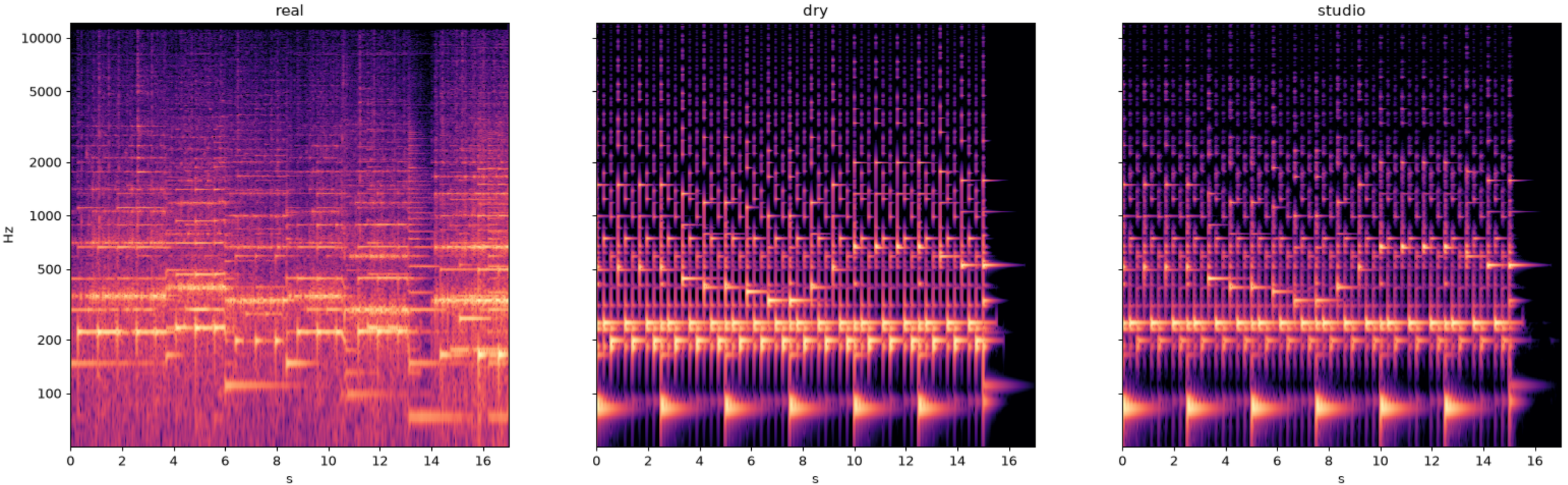
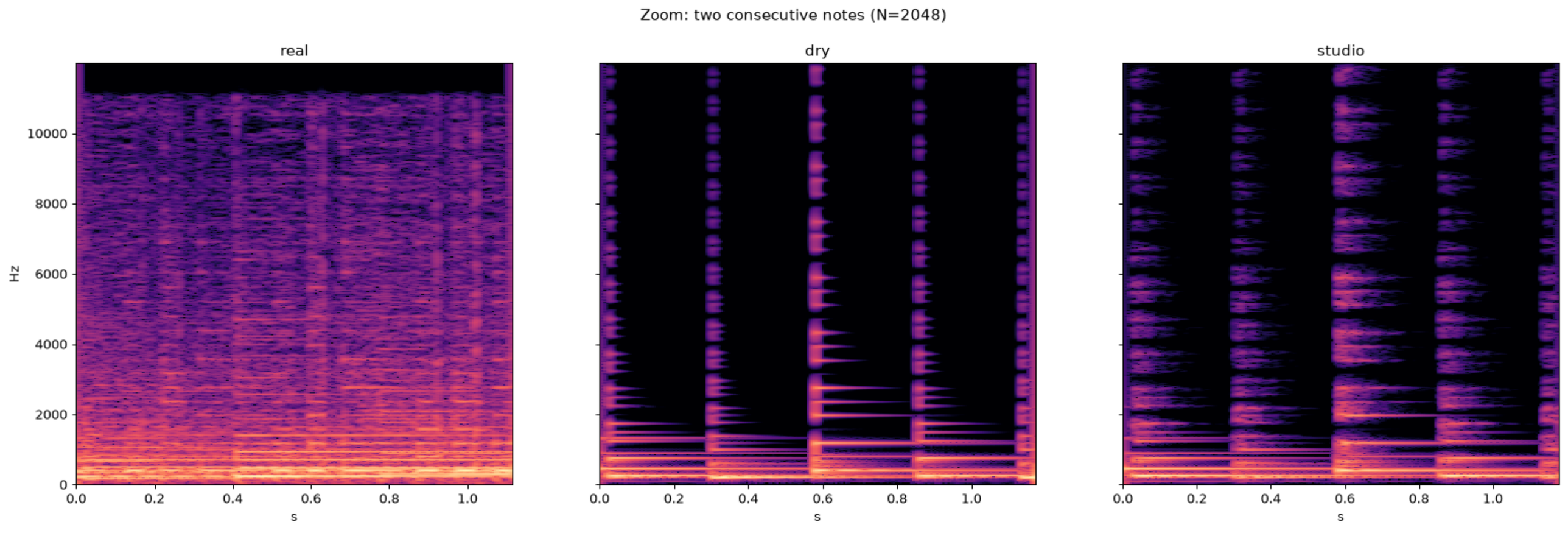
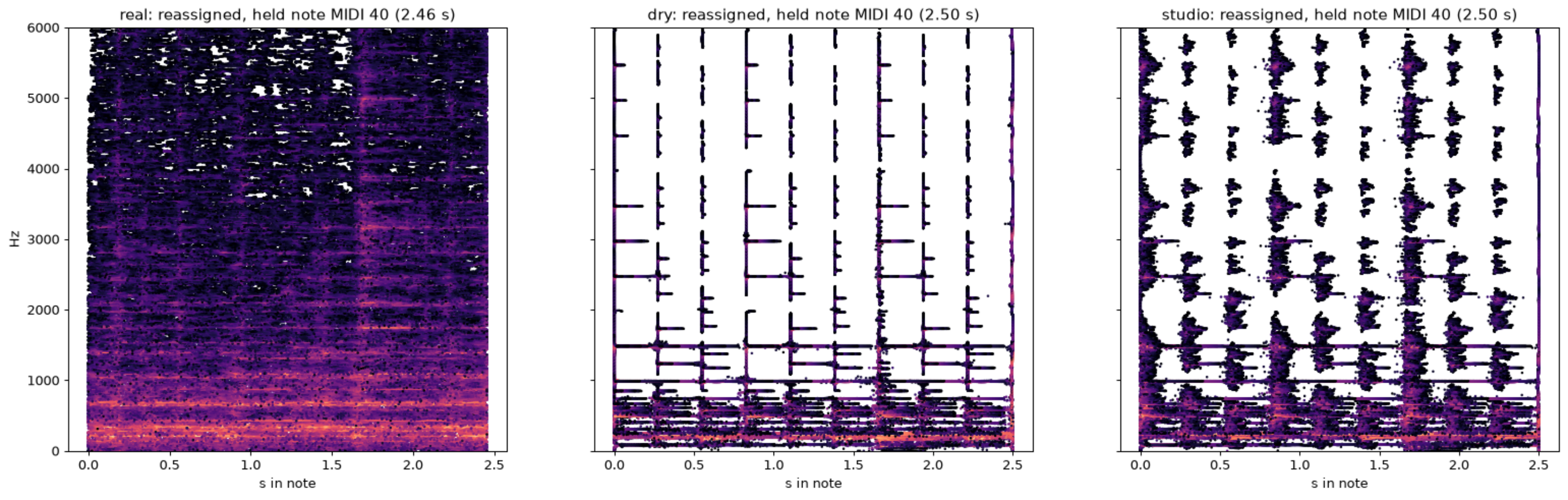

real: waveform, RMS and peak envelope (equal RMS)
RMS x sqrt2
peak
dry: waveform, RMS and peak envelope (equal RMS)
RMS x sqrt2
peak
studio: waveform, RMS and peak envelope (equal RMS)
RMS x sqrt2
peak
0.5
0.0
−0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
s

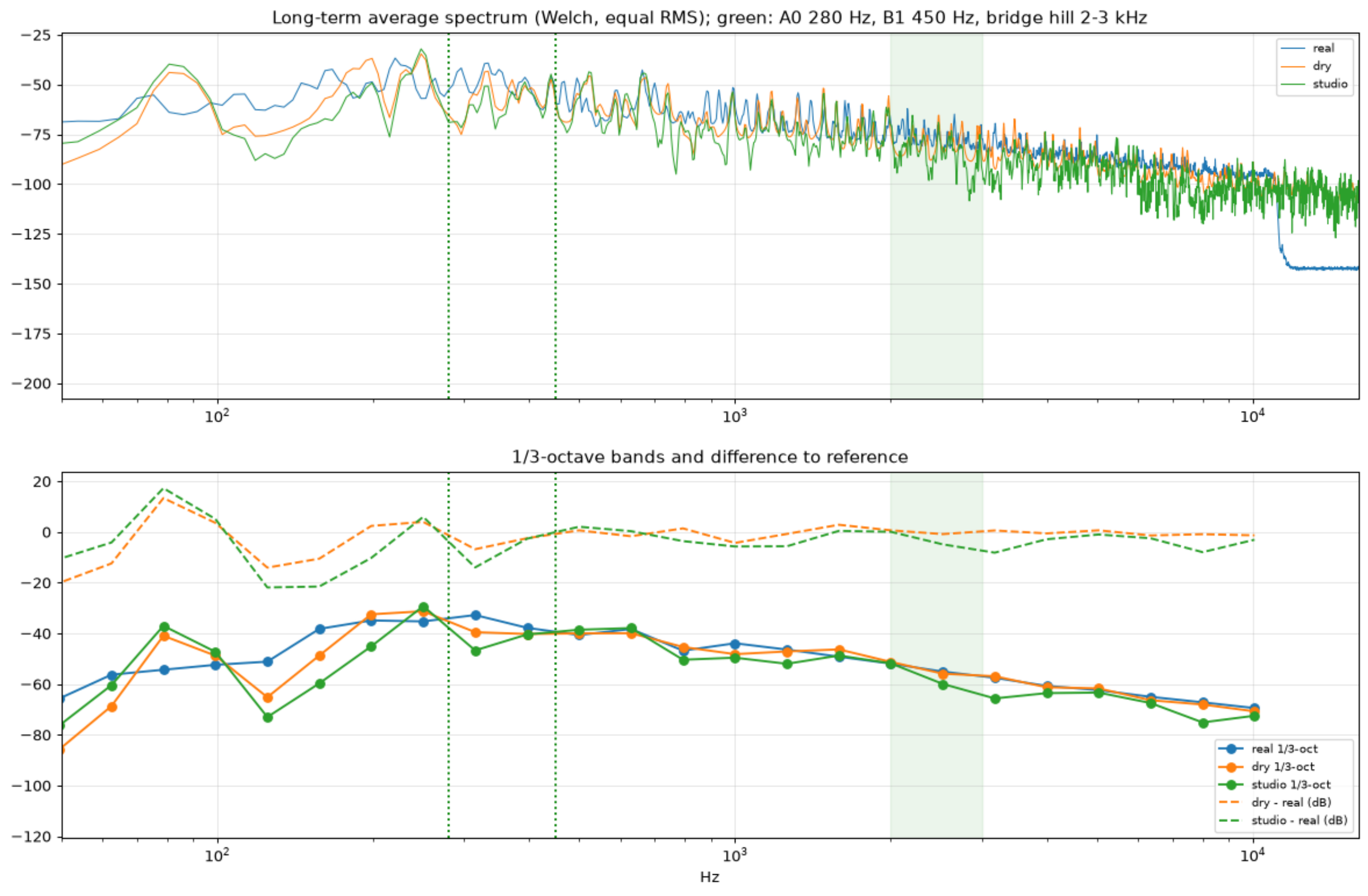

Momentary loudness (400 ms, BS.1770 K-weighted)
LUFS
−20
−40
−60
−80
real
dry
studio
Short-term loudness (3 s)
LUFS
−19
−20
−21
−22
−23
−24
real
dry
studio
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
s

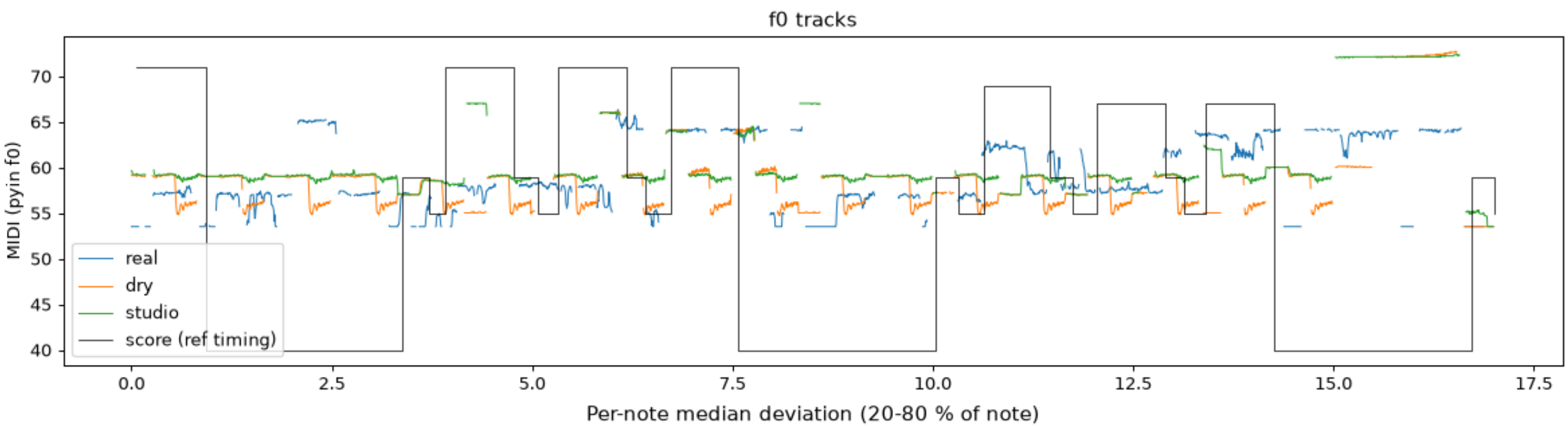

Linear spectrogram 0-12 kHz (N=8192, dB re clip max)
real
dry
studio
Hz
10000
8000
6000
4000
2000
0
0
2
4
6
8
10
12
14
16
s

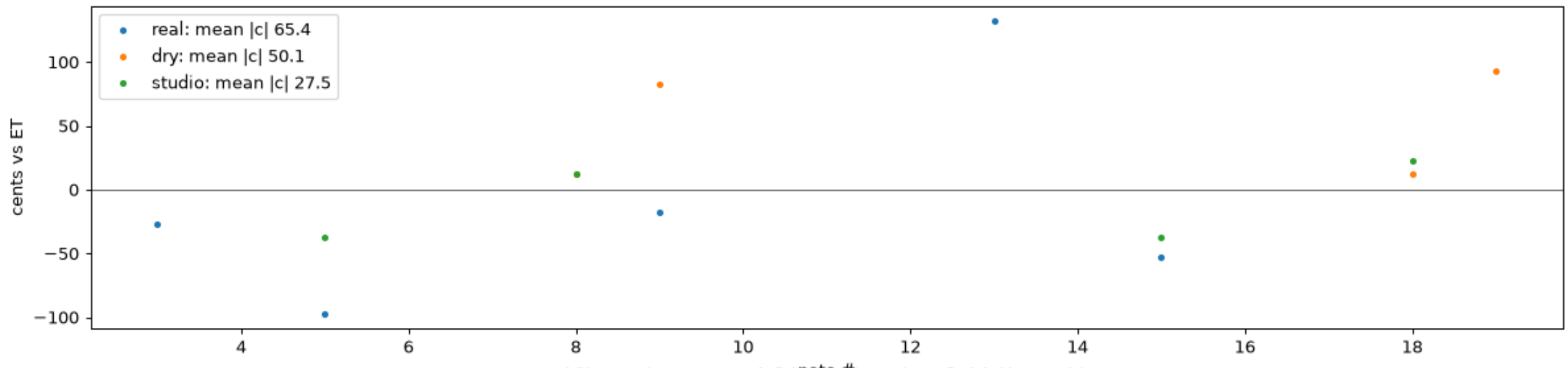

Log-frequency spectrogram 50 Hz-12 kHz
real
dry
studio
Hz
10000
5000
2000
1000
500
200
100
0
2
4
6
8
10
12
14
16
s

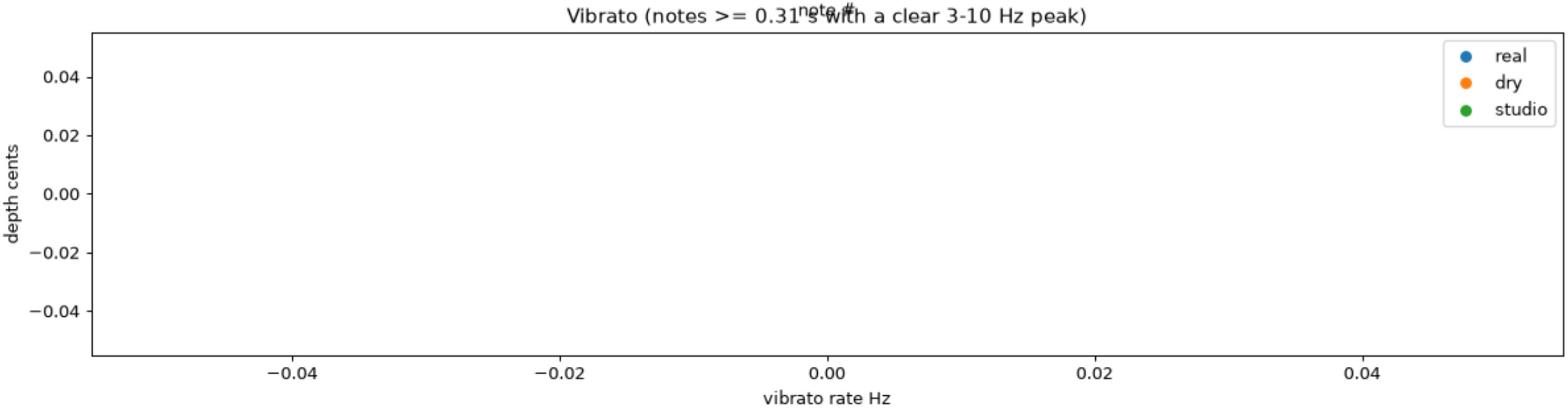

Zoom: two consecutive notes (N=2048)
real
dry
studio
Hz
10000
8000
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
s

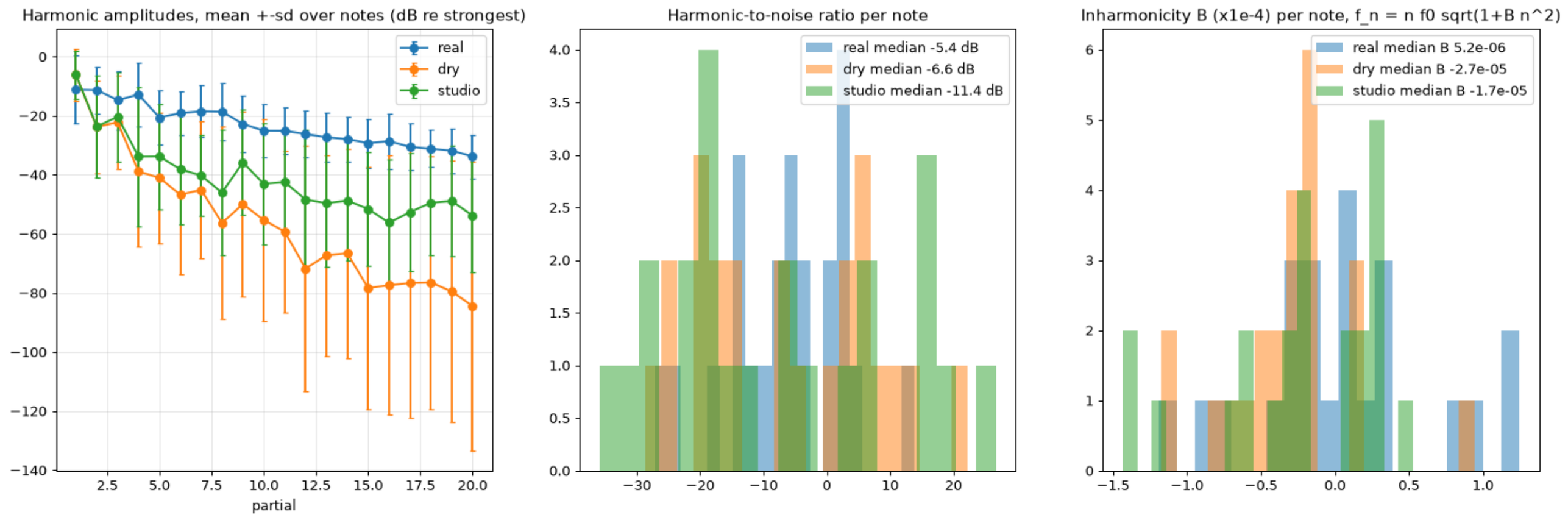

real: reassigned, held note MIDI 40 (2.46 s)
dry: reassigned, held note MIDI 40 (2.50 s)
studio: reassigned, held note MIDI 40 (2.50 s)
Hz
0
1000
2000
3000
4000
5000
6000
0.0
0.5
1.0
1.5
2.0
2.5
s in note


Long-term average spectrum (Welch, equal RMS); green: A0 280 Hz, B1 450 Hz, bridge hill 2-3 kHz
real
dry
studio
−25
−50
−75
−100
−125
−150
−175
−200
10^2
10^3
10^4
1/3-octave bands and difference to reference
real 1/3-oct
dry 1/3-oct
studio 1/3-oct
dry - real (dB)
studio - real (dB)
20
0
−20
−40
−60
−80
−100
−120
10^2
10^3
10^4
Hz


f0 tracks
MIDI (pyin f0)
70
65
60
55
50
45
40
real
dry
studio
score (ref timing)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Per-note median deviation (20-80 % of note)
real: mean |c| 65.4
dry: mean |c| 50.1
studio: mean |c| 27.5
cents vs ET
100
50
0
−50
−100
4
6
8
10
12
14
16
18
note #


Vibrato (notes >= 0.31 s with a clear 3-10 Hz peak)
depth cents
0.04
0.02
0.00
−0.02
−0.04
real
dry
studio
−0.04
−0.02
0.00
0.02
0.04
vibrato rate Hz


Harmonic amplitudes, mean +-sd over notes (dB re strongest)
real
dry
studio
0
−20
−40
−60
−80
−100
−120
−140
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
partial
Harmonic-to-noise ratio per note
real median -5.4 dB
dry median -6.6 dB
studio median -11.4 dB
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−30
−20
−10
0
10
20
Inharmonicity B (x1e-4) per note, f_n = n f0 sqrt(1+B n^2)
real median B 5.2e-06
dry median B -2.7e-05
studio median B -1.7e-05
6
5
4
3
2
1
0
−1.5
−1.0
−0.5
0.0
0.5
1.0

Spectral descriptors over time

centroid

real median 1.07e+03
dry median 384
studio median 519

rolloff

real median 1.7e+03
dry median 538
studio median 603

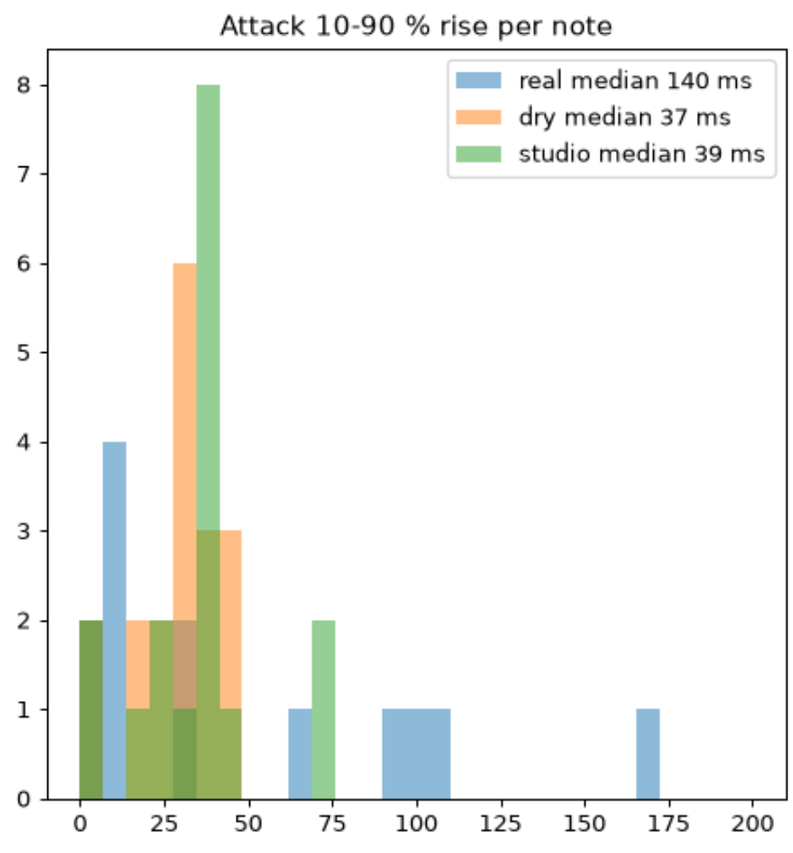
flatness

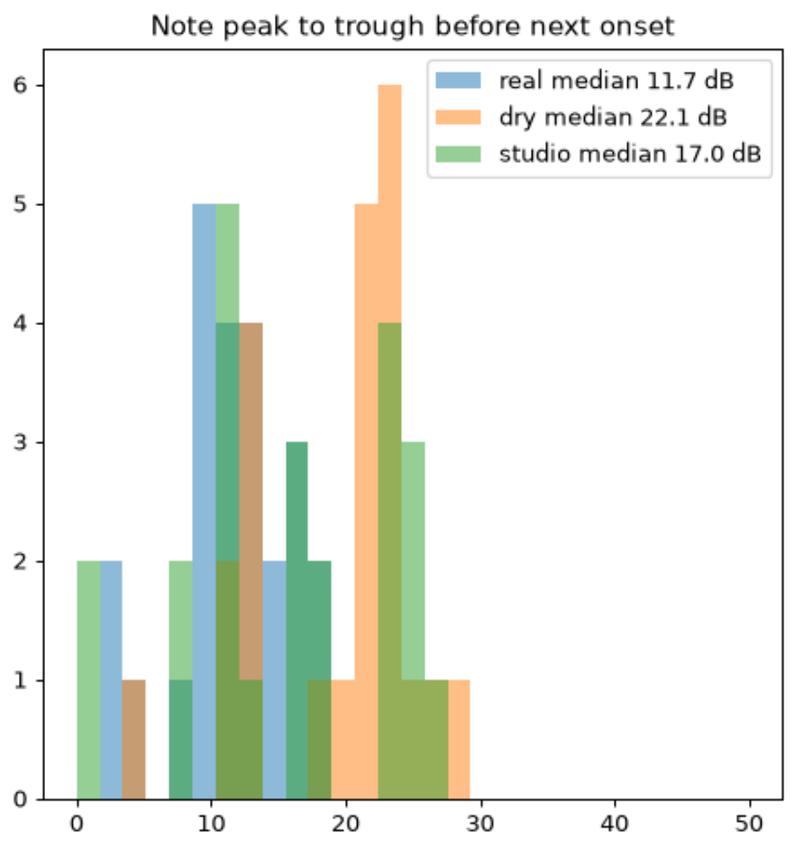
real median 4.04e-06
dry median 2.29e-10
studio median 9.04e-08

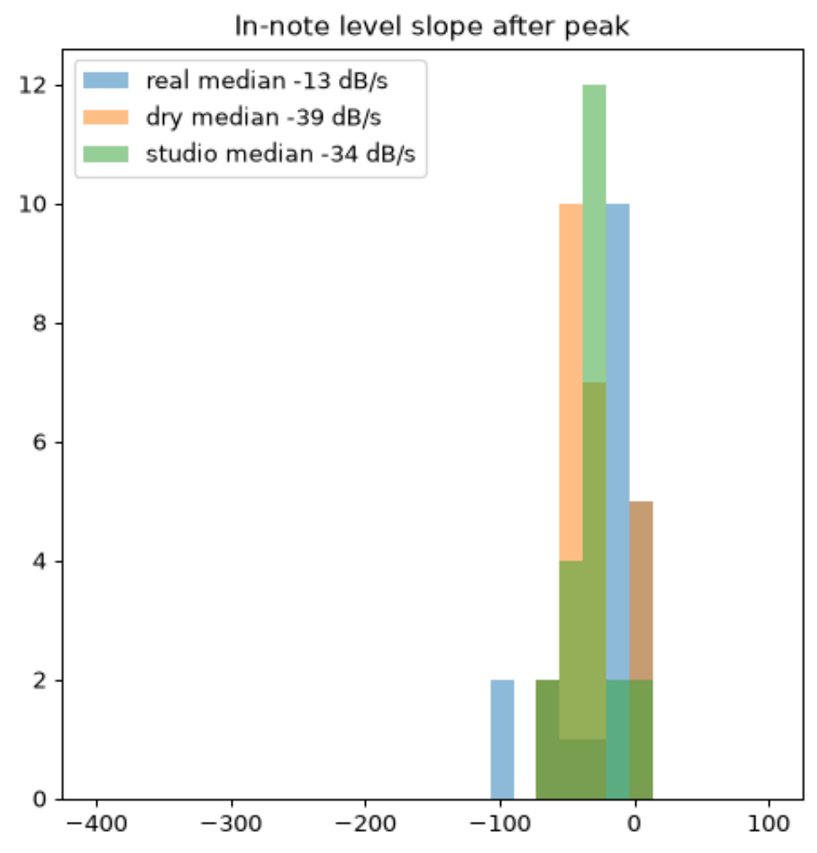
flux

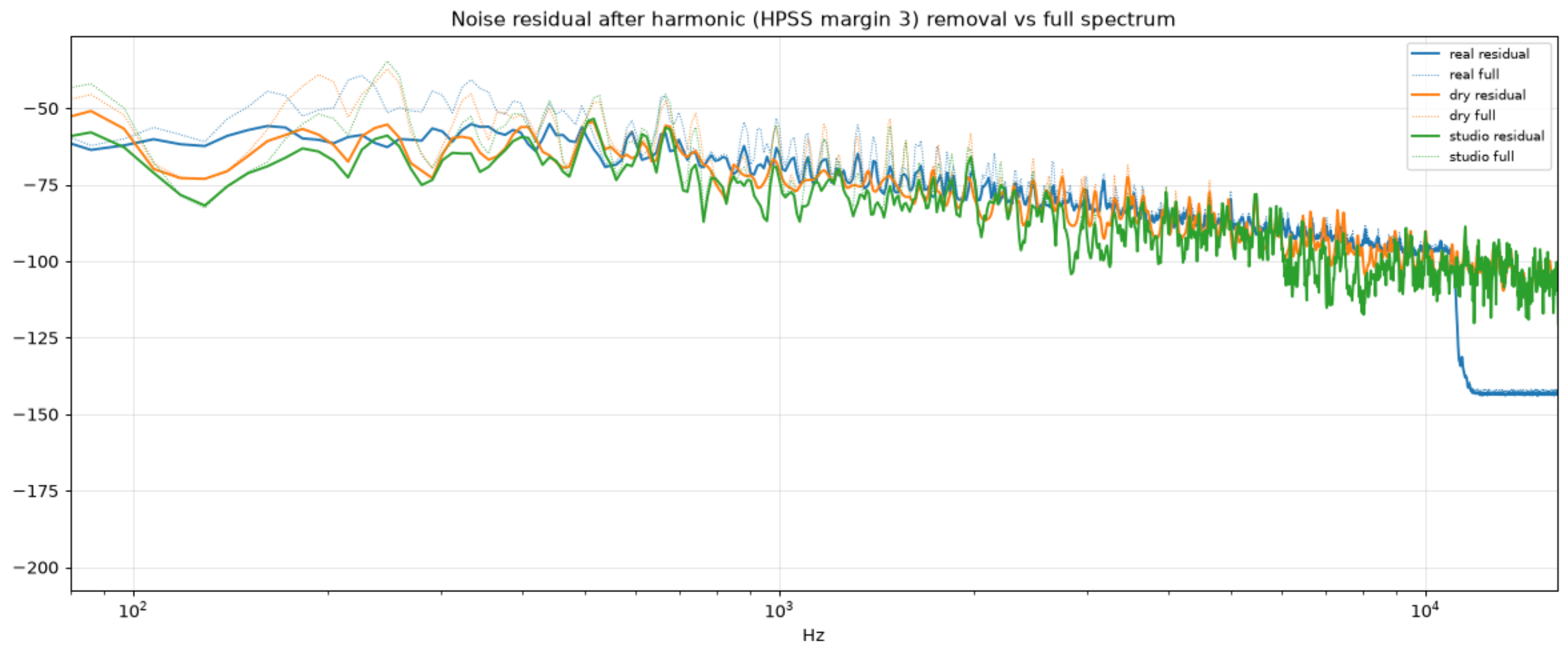
real median 0.97
dry median 0.0333
studio median 0.0662

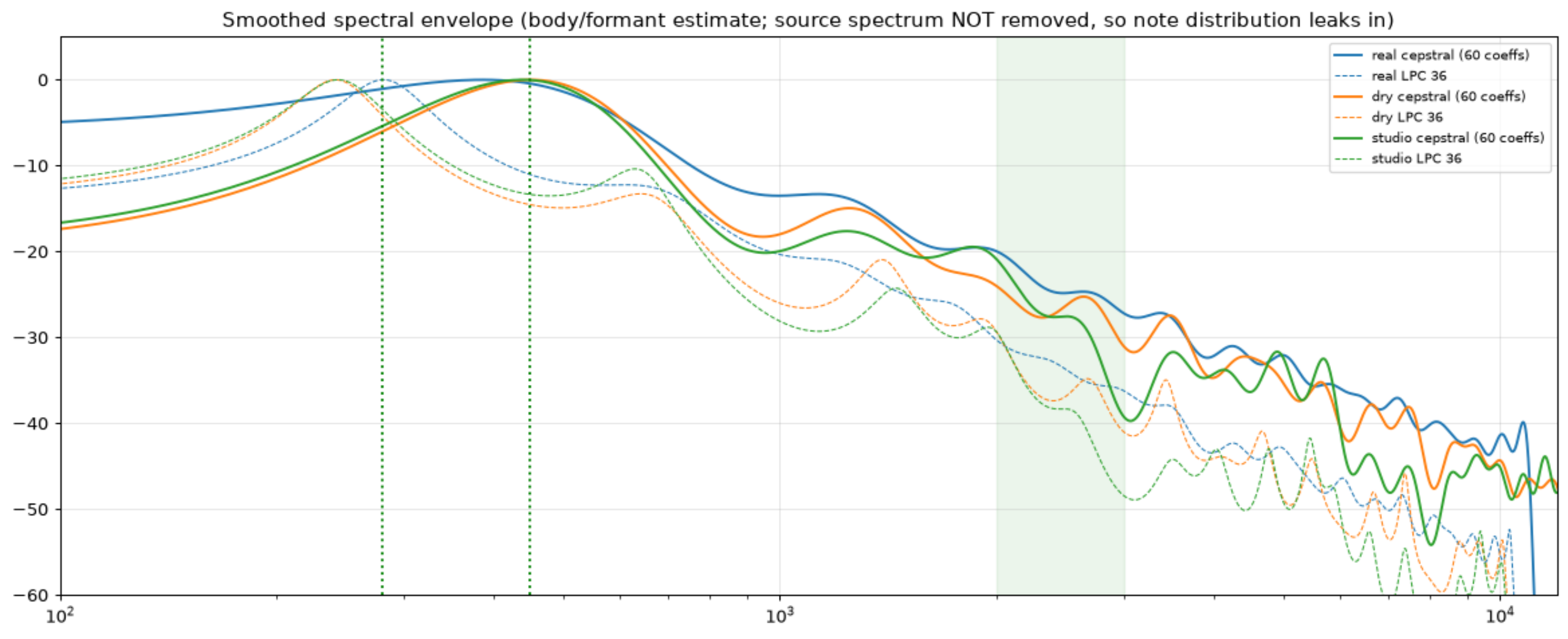
s

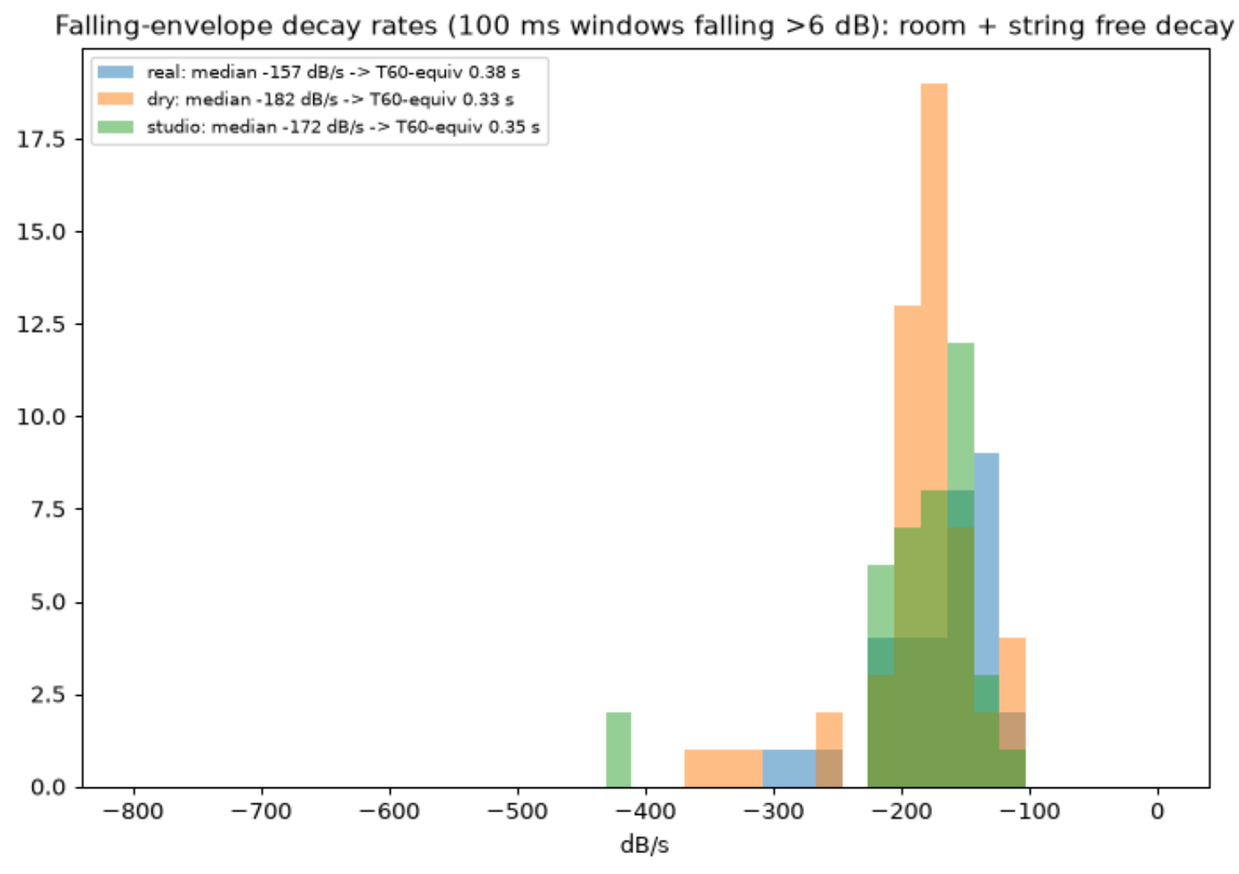

Attack 10-90 % rise per note
real median 140 ms
dry median 37 ms
studio median 39 ms
0
1
2
3
4
5
6
7
8
0
25
50
75
100
125
150
175
200

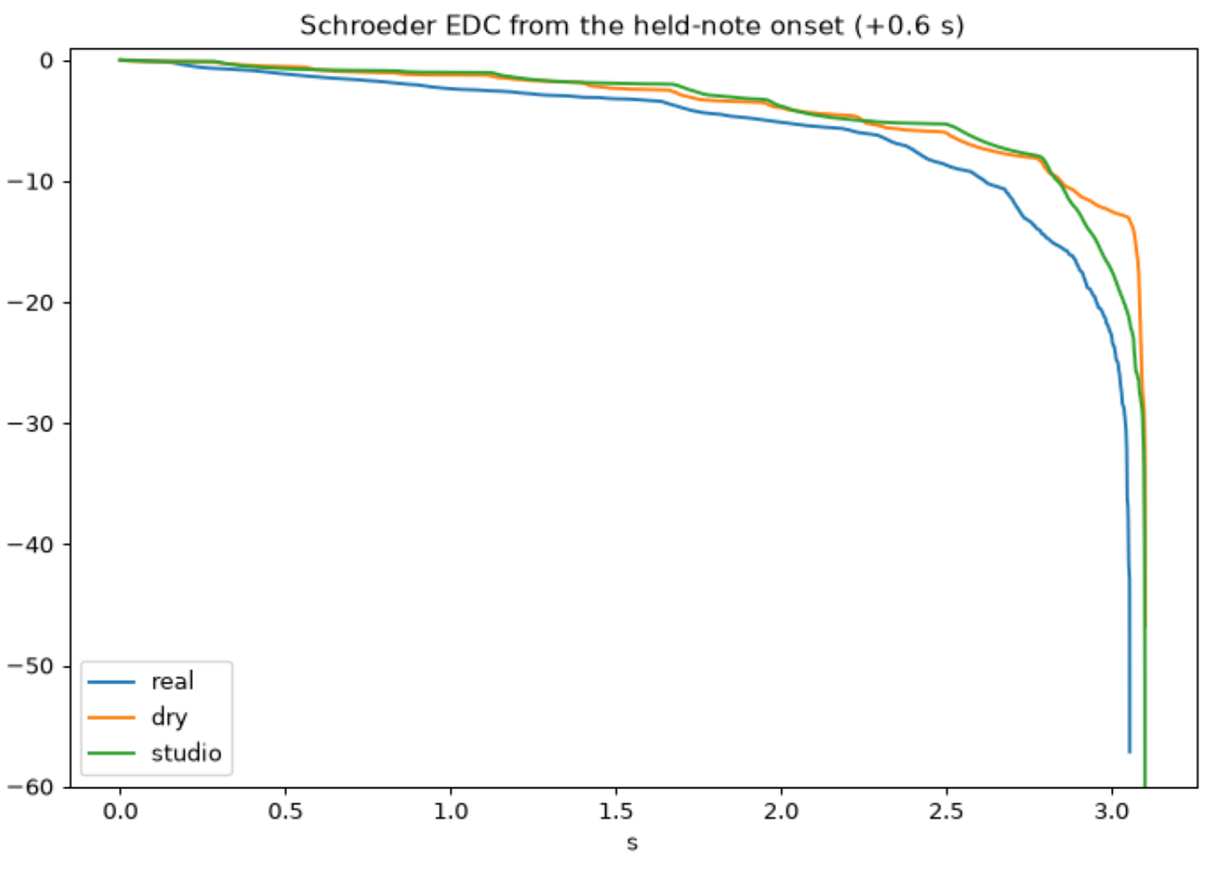

Note peak to trough before next onset
real median 11.7 dB
dry median 22.1 dB
studio median 17.0 dB
0
1
2
3
4
5
6
0
10
20
30
40
50

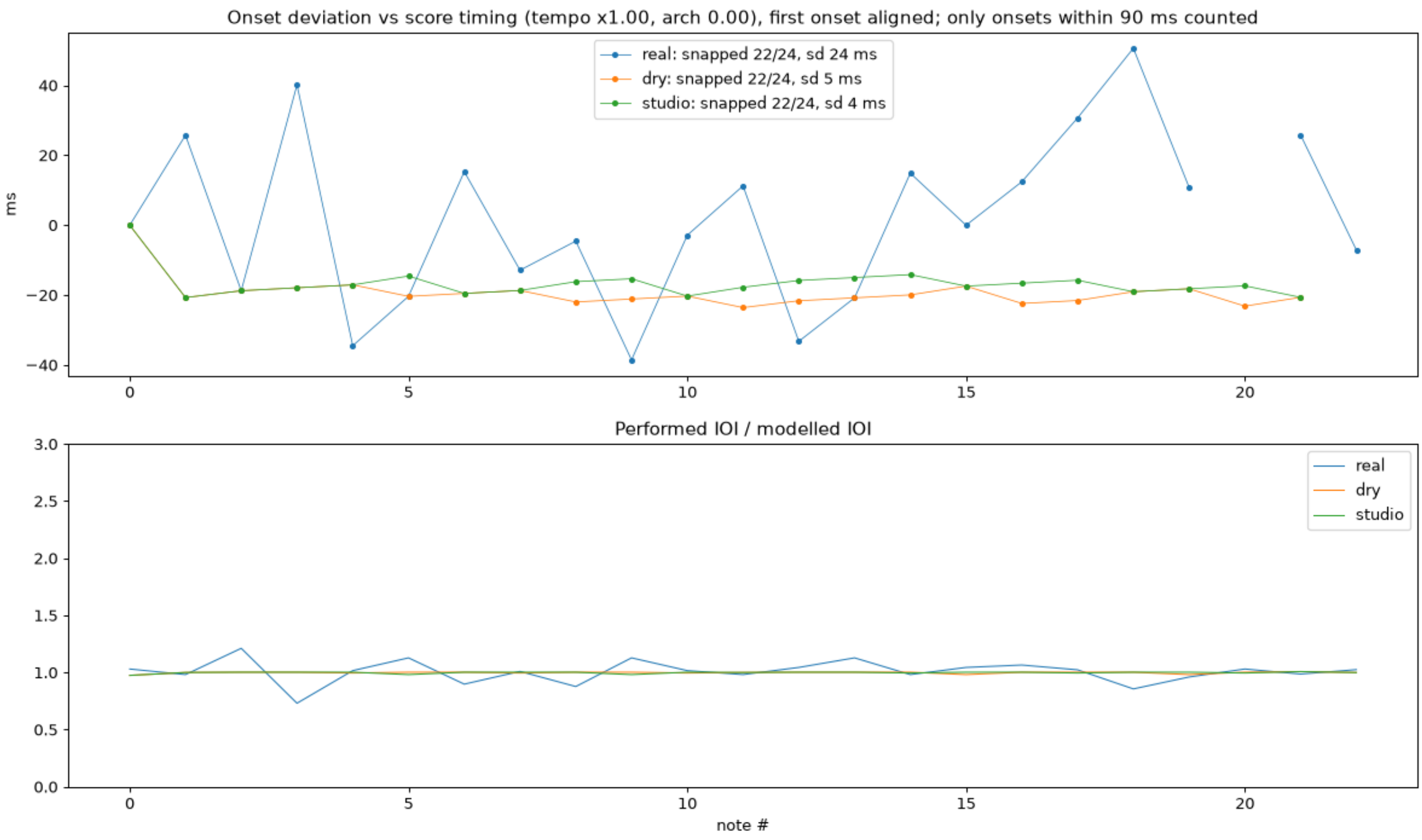

In-note level slope after peak
real median -13 dB/s
dry median -39 dB/s
studio median -34 dB/s
0
2
4
6
8
10
12
−400
−300
−200
−100
0
100


Noise residual after harmonic (HPSS margin 3) removal vs full spectrum
real residual
real full
dry residual
dry full
studio residual
studio full
−50
−75
−100
−125
−150
−175
−200
10^2
10^3
10^4
Hz


Smoothed spectral envelope (body/formant estimate; source spectrum NOT removed, so note distribution leaks in)
real cepstral (60 coeffs)
real LPC 36
dry cepstral (60 coeffs)
dry LPC 36
studio cepstral (60 coeffs)
studio LPC 36
0
−10
−20
−30
−40
−50
−60
10^2
10^3
10^4


Falling-envelope decay rates (100 ms windows falling >6 dB): room + string free decay
real: median -157 dB/s -> T60-equiv 0.38 s
dry: median -182 dB/s -> T60-equiv 0.33 s
studio: median -172 dB/s -> T60-equiv 0.35 s
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−800
−700
−600
−500
−400
−300
−200
−100
0
dB/s


Schroeder EDC from the held-note onset (+0.6 s)
0
−10
−20
−30
−40
−50
−60
real
dry
studio
0.0
0.5
1.0
1.5
2.0
2.5
3.0
s


Onset deviation vs score timing (tempo x1.00, arch 0.00), first onset aligned; only onsets within 90 ms counted
real: snapped 22/24, sd 24 ms
dry: snapped 22/24, sd 5 ms
studio: snapped 22/24, sd 4 ms
ms
40
20
0
−20
−40
0
5
10
15
20
Performed IOI / modelled IOI
real
dry
studio
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
5
10
15
20
note #

Scorecard vs real (page 1)

metric	clip	ours	real	diff	verdict	likely physical cause in model
Momentary loudness range p95-p10 (LU)	dry	4.98	8.07	-3.09	close	no phrase-level dynamics (velocity only 81/101); room stage wet level
Momentary loudness range p95-p10 (LU)	studio	7.32	8.07	-0.758	match	
Short-term loudness std (LU)	dry	0.704	1.13	-0.426	match	
Short-term loudness std (LU)	studio	0.676	1.13	-0.454	match	
RMS envelope swing p90-p10 (dB)	dry	18.9	10.1	+8.81	close	spiccato lift + rest damping vs reverb fill
RMS envelope swing p90-p10 (dB)	studio	19.4	10.1	+9.29	close	spiccato lift + rest damping vs reverb fill
Band 400-500 Hz re 0.5-1.5k (dB)	dry	3.27	6.64	-3.37	close	no violin body (B1-/B1+ corpus modes)
Band 400-500 Hz re 0.5-1.5k (dB)	studio	3.37	6.64	-3.27	close	no violin body (B1-/B1+ corpus modes)
Band 2-3 kHz re 0.5-1.5k (dB)	dry	-18.6	-15.9	-2.75	close	no bridge rocking mode (bridge hill)
Band 2-3 kHz re 0.5-1.5k (dB)	studio	-21.3	-15.9	-5.47	close	no bridge rocking mode (bridge hill)
Band 5-10 kHz re 0.5-1.5k (dB)	dry	-33.1	-31.7	-1.39	match	
Band 5-10 kHz re 0.5-1.5k (dB)	studio	-35.2	-31.7	-3.45	close	dry: no body/air/mic rolloff; room IR supplies it
Band 200-320 Hz re 0.5-1.5k (dB)	dry	13.9	13.9	+0.0228	match	
Band 200-320 Hz re 0.5-1.5k (dB)	studio	14.9	13.9	+1.01	match	
Max |1/3-oct diff| 100 Hz-10 kHz (dB)	dry	14	0	+14	wrong	body transfer function missing
Max |1/3-oct diff| 100 Hz-10 kHz (dB)	studio	21.8	0	+21.8	wrong	body transfer function missing
Pitch mean |cents| vs ET (pyin)	dry	50.1	65.4	-15.4	wrong	smooth-friction sharp offset; player intonation
Pitch mean |cents| vs ET (pyin)	studio	27.5	65.4	-37.9	wrong	smooth-friction sharp offset; player intonation
Note-centre pitch mean |cents| (v10: mid-50pct, PV-re	dry	4.86	nan	+nan	n/a	friction-curve offset + seeded intonation (4 c); real: player intonation
Note-centre pitch mean |cents| (v10: mid-50pct, PV-re	studio	7.78	nan	+nan	n/a	friction-curve offset + seeded intonation (4 c); real: player intonation
Note-centre mean |cents|, notes >= 0.25 s (v10)	dry	4.86	nan	+nan	n/a	
Note-centre mean |cents|, notes >= 0.25 s (v10)	studio	7.78	nan	+nan	n/a	

Scorecard vs real (page 2)

metric	clip	ours	real	diff	verdict	likely physical cause in model
Note-centre mean |cents|, notes < 0.25 s (v10)	dry	nan	nan	+nan	n/a	
Note-centre mean |cents|, notes < 0.25 s (v10)	studio	nan	nan	+nan	n/a	
Note-centre measurable notes (count)	dry	1	0	+1	match	
Note-centre measurable notes (count)	studio	4	0	+4	wrong	
Note-centre pitch mean signed cents	dry	4.86	nan	+nan	n/a	smooth-friction sharp offset
Note-centre pitch mean signed cents	studio	-0.371	nan	+nan	n/a	smooth-friction sharp offset
Note-centre |cents| re own median tuning	dry	0	nan	+nan	n/a	intonation scatter (ours seeded 4 c; real player)
Note-centre |cents| re own median tuning	studio	7.78	nan	+nan	n/a	intonation scatter (ours seeded 4 c; real player)
Own median tuning offset (cents)	dry	4.86	nan	+nan	n/a	reference pitch (real may tune above A440)
Own median tuning offset (cents)	studio	-0.219	nan	+nan	n/a	reference pitch (real may tune above A440)
Pitch spread std cents	dry	37.7	77.6	-39.9	wrong	spiccato notes start in weak/non-Helmholtz contact
Pitch spread std cents	studio	27.7	77.6	-49.9	wrong	spiccato notes start in weak/non-Helmholtz contact
Vibrato rate Hz (median, measurable notes)	dry	nan	nan	+nan	n/a	fixed 6 Hz vibrato with 0.3 s onset
Vibrato rate Hz (median, measurable notes)	studio	nan	nan	+nan	n/a	fixed 6 Hz vibrato with 0.3 s onset
Vibrato depth cents	dry	nan	nan	+nan	n/a	fixed vibrato depth; short notes never reach full depth
Vibrato depth cents	studio	nan	nan	+nan	n/a	fixed vibrato depth; short notes never reach full depth
Harmonic slope partials 1-12 (dB/partial)	dry	-4.79	-1.42	-3.38	wrong	ideal sawtooth Helmholtz source, no body filter, bow-position comb
Harmonic slope partials 1-12 (dB/partial)	studio	-2.94	-1.42	-1.53	close	ideal sawtooth Helmholtz source, no body filter, bow-position comb
Partial 2 re strongest (dB)	dry	-23.8	-11.3	-12.5	wrong	body/radiation weighting absent
Partial 2 re strongest (dB)	studio	-23.6	-11.3	-12.3	wrong	body/radiation weighting absent
HNR median (dB)	dry	-6.56	-5.42	-1.14	match	
HNR median (dB)	studio	-11.4	-5.42	-6	close	stick-slip/bow-change transients; no bow-width/torsion smoothing

Scorecard vs real (page 3)

metric	clip	ours	real	diff	verdict	likely physical cause in model
Inharmonicity B median (x1e-4)	dry	-0.273	0.0518	-0.324	match	
Inharmonicity B median (x1e-4)	studio	-0.166	0.0518	-0.218	match	
Spectral centroid median	dry	384	1.07e+03	-682	wrong	source brightness vs body/room filtering
Spectral centroid median	studio	519	1.07e+03	-547	close	source brightness vs body/room filtering
Spectral rolloff median	dry	538	1.7e+03	-1.16e+03	close	source brightness vs body/room filtering
Spectral rolloff median	studio	603	1.7e+03	-1.1e+03	close	source brightness vs body/room filtering
Spectral flatness median	dry	2.29e-10	4.04e-06	-4.04e-06	match	
Spectral flatness median	studio	9.04e-08	4.04e-06	-3.95e-06	match	
Spectral flux median	dry	0.0333	0.97	-0.937	wrong	articulation density
Spectral flux median	studio	0.0662	0.97	-0.904	wrong	articulation density
Attack rise 10-90% median (ms)	dry	37	140	-103	wrong	bow landing ramps (8-30 ms) vs real bow + room smear
Attack rise 10-90% median (ms)	studio	39.2	140	-101	wrong	bow landing ramps (8-30 ms) vs real bow + room smear
Note peak-to-gap depth median (dB)	dry	22.1	11.7	+10.4	wrong	rest damping / reverb fill
Note peak-to-gap depth median (dB)	studio	17	11.7	+5.26	close	rest damping / reverb fill
In-note decay slope median (dB/s)	dry	-39	-13	-25.9	close	bounce half-sine envelope, string loss
In-note decay slope median (dB/s)	studio	-34.1	-13	-21.1	close	bounce half-sine envelope, string loss
Harmonic/residual energy HPSS (dB)	dry	9.94	8.66	+1.28	match	
Harmonic/residual energy HPSS (dB)	studio	12.9	8.66	+4.26	close	transient noise; HPSS counts onsets as noise
Envelope peak near 400-550 Hz present	dry	1	1	+0	match	
Envelope peak near 400-550 Hz present	studio	1	1	+0	match	
Envelope peak near 2-3 kHz present	dry	1	0	+1	wrong	no bridge hill
Envelope peak near 2-3 kHz present	studio	0	0	+0	match	

Scorecard vs real (page 4)

metric	clip	ours	real	diff	verdict	likely physical cause in model
Decay T60-equiv from falling envelopes (s)	dry	0.329	0.383	-0.0537	match	
Decay T60-equiv from falling envelopes (s)	studio	0.348	0.383	-0.0343	match	
Onset deviation sd vs model timing (ms)	dry	4.56	23.8	-19.2	close	rule-based timing vs real rubato
Onset deviation sd vs model timing (ms)	studio	4.09	23.8	-19.7	close	rule-based timing vs real rubato
Onsets detected per s	dry	3.24	11.1	-7.82	wrong	articulation clarity
Onsets detected per s	studio	3.24	11.1	-7.82	wrong	articulation clarity